ECO3121

Mid-term Review

Fall 2022

Main topics of our lecture

1. Statistic review
2. Simple regression
3. Multiple regression

Statistic Review

1. **Causality:** Experimental data and theoretical modeling
2. **Probability:** Marginal, joint and conditional probability(Bayes' Rule and independent event)
3. **Random variable:** Discrete and continuous variable,
   Expected value
   Variance and standard deviation
   Covariance and correlation
   Conditional Expectation
4. **Random distribution:** Pdf and Cdf

Simple Regression

1. Definition of the Simple Regression Model
2. Deriving the Ordinary Least Squares Estimates
3. Properties of OLS on any Sample of Data
4. Units of Measurement and Functional Form
5. Testing
6. Confidence interval

Definition of the Simple Regression Model

Assumptions:

Assumption SLR.1 (Linear in Parameters)
In the population model, the dependent variable, y, is related to the independent variable, x, and the error (or disturbance), u, as

$$y = \beta_0 + \beta_1 x + u,$$

where β_0 and β_1 are the population intercept and slope parameters, respectively.

Assumption SLR.2 (Random Sampling)
We have a random sample of size n, $\{(x_i,y_i): i = 1, 2, \ldots, n\}$, following the population model in Assumption SLR.1.

Assumption SLR.3 (Sample Variation in the Explanatory Variable)
The sample outcomes on x, namely, $\{x_i, i = 1, \ldots, n\}$, are not all the same value.

Assumption SLR.4 (Zero Conditional Mean)
The error u has an expected value of zero given any value of the explanatory variable. In other words,

$$E(u|x) = 0.$$

Assumption SLR.5 (Homoskedasticity)
The error u has the same variance given any value of the explanatory variable. In other words,

$$\text{Var}(u|x) = \sigma^2.$$

Pros

1. Unbiased
2. Efficient

Deriving the Ordinary Least Squares Estimate: Least Sum of Squared Error (SSE)

- **Deriving the OLS estimate**
- Where does the name "**Ordinary Least Squares**" come from?
  For any candidates $\hat{\beta}_0$ and $\hat{\beta}_1$, define a **fitted (or predicted) value** for each data point i as:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

  We have n of these. It is the value we predict for y_i given that x has taken on the value x_i .

- The mistake we make is the residual:

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i,$$

  and we have n residuals.

Deriving the Ordinary Least Squares Estimate: Least Sum of Squared Error (SSE)

Deriving the Ordinary Least Squares Estimate

- **What does "as good as possible" mean?**
- Regression residuals

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

- Choose $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize sum of squared regression residuals

$$\min \sum_{i=1}^{n} \hat{u}_i^2 \quad \rightarrow \min \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

- First order conditions

$$n^{-1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$n^{-1} \sum_{i=1}^{n} x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

Deriving the Ordinary Least Squares Estimate: Least Sum of Squared Error (SSE)

Deriving the Ordinary Least Squares Estimate

- **First equation**

$$
\begin{aligned}
n^{-1}\sum_{i=1}^{n}(y_i-\hat{\beta}_0-\hat{\beta}_1x_i) &= n^{-1}\sum_{i=1}^{n}y_i-n^{-1}\sum_{i=1}^{n}\hat{\beta}_0-n^{-1}\sum_{i=1}^{n}\hat{\beta}_1x_i \\
&= n^{-1}\sum_{i=1}^{n}y_i-\hat{\beta}_0-\hat{\beta}_1\left(n^{-1}\sum_{i=1}^{n}x_i\right) \\
&= \bar{y}-\hat{\beta}_0-\hat{\beta}_1\bar{x}=0
\end{aligned}
$$

implies

$$\bar{y}=\hat{\beta}_0+\hat{\beta}_1\bar{x}$$

and plug this into the **second equation**

$$\sum_{i=1}^{n}x_i[y_i-(\bar{y}-\hat{\beta}_1\bar{x})-\hat{\beta}_1x_i]=0$$

Deriving the Ordinary Least Squares Estimate: Least Sum of Squared Error (SSE)

Deriving the Ordinary Least Squares Estimate

Simple algebra gives

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \hat{\beta}_1 \left[\sum_{i=1}^{n} x_i(x_i - \bar{x}) \right] \qquad (1)$$

Because

$$\sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

$$\sum_{i=1}^{n} x_i(y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} (x_i - \bar{x}) y_i$$

$$\sum_{i=1}^{n} x_i(x_i - \bar{x}) = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

one can use equation (1) to derive:

- **Ordinary Least Squares (OLS) estimates**

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Interpretations of Regression Results

1. Also remember we can only draw correlations rather than causation effect from observational data
2. Units of Measurement and Functional Form (with log or without log)

Model	Dep. Var.	Indep. Var.	β_1
Level-Level	y	x	$\Delta y = \beta_1 \Delta x$
Level-Log	y	$\log(x)$	$\Delta y = (\beta_1/100)\%\Delta x$
Log-Level	$\log(y)$	x	$\%\Delta y = (100\beta_1)\Delta x$
Log-Log	$\log(y)$	$\log(x)$	$\%\Delta y = \beta_1 \%\Delta x$

Properties of OLS Estimators

1. Unbiasedness: Expected value of beta_1_hat = beta 1

$$
\begin{aligned}
\sum_{i=1}^{n}(x_i-\bar{x})y_i &= \sum_{i=1}^{n}(x_i-\bar{x})(\beta_0+\beta_1 x_i+u_i) \\
&= \beta_0\sum_{i=1}^{n}(x_i-\bar{x})+\beta_1\sum_{i=1}^{n}(x_i-\bar{x})x_i+\sum_{i=1}^{n}(x_i-\bar{x})u_i \\
&= 0+\beta_1\sum_{i=1}^{n}(x_i-\bar{x})^2+\sum_{i=1}^{n}(x_i-\bar{x})u_i \\
&= \beta_1 SST_x+\sum_{i=1}^{n}(x_i-\bar{x})u_i
\end{aligned}
$$

Properties of OLS Estimators

- Because we assume SLR.4, that is, $E(u|x) = 0$, whenever we assume SLR.5, we can also write

$$E(u^2|x) = \sigma^2 = E(u^2)$$

- Under the population Assumptions SLR.1 $(y = \beta_0 + \beta_1 x + u)$, SRL.4 $(E(u|x) = 0)$ and SLR.5 $(Var(u|x) = \sigma^2)$,

$$\begin{aligned} E(y|x) &= \beta_0 + \beta_1 x \\ Var(y|x) &= \sigma^2 \end{aligned}$$

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2}{SST_x}$$

$$Var(\hat{\beta}_0) = \frac{\sigma^2 n^{-1} \sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\sigma^2 n^{-1} \sum_{i=1}^{n} x_i^2}{SST_x}$$

Testing

- Theorem (t-distribution for the standardized estimators)

Under assumptions SLR.1 – SLR.6:

$$\frac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)} \sim t_{n-k-1}$$

If the standardization is done using the estimated standard deviation (= standard error), the normal distribution is replaced by a t-distribution

Note: The t-distribution is close to the standard normal distribution if n-k-1 is large.

- t-statistic (or t-ratio)

$$t_{\hat{\beta}_j} \equiv \frac{\hat{\beta}_j}{se(\hat{\beta}_j)}$$

The t-statistic will be used to test the above null hypothesis. The farther the estimated coefficient is away from zero, the less likely it is that the null hypothesis holds true. But what does "far" away from zero mean?

This depends on the variability of the estimated coefficient, i.e. its standard deviation. The t-statistic measures how many estimated standard deviations the estimated coefficient is away from zero.

- Distribution of the t-statistic if the null hypothesis is true

$$t_{\hat{\beta}_j} \equiv \hat{\beta}_j / se(\hat{\beta}_j) = (\hat{\beta}_j - \beta_j)/se(\hat{\beta}_j) \sim t_{n-k-1}$$

- Goal: Define a rejection rule so that, if it is true, H0 is rejected only with a small probability (= significance level, e.g. 5%)

Multiple Regression

1. Motivation for multiple Regression(omitted variable biasedness)
2. Mechanics and interpretation of OLS
3. The expected value of the OLS Estimators

Mechanics and Interpretation of OLS

Gauss Markov assumptions in MLR:

- MLR.1: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + u$
- MLR.2: random sampling from the population
- MLR.3: no perfect collinearity in the sample
- MLR.4: $E(u|x_1, ..., x_k) = E(u) = 0$ (exogenous explanatory variables)
- MLR.5: $Var(u|x_1, ..., x_k) = Var(u) = \sigma^2$ (homoskedasticity)

Mechanics and Interpretation of OLS

1. **How to derive the coefficient:** still the Least sum of squared error
   - Can be solved with multivariable calculus. The **OLS first order conditions** are the $k+1$ linear equations in the $k+1$ unknowns $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_k$

$$
\begin{aligned}
\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \ldots - \hat{\beta}_k x_{ik}) &= 0 \\
\sum_{i=1}^{n} x_{i1}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \ldots - \hat{\beta}_k x_{ik}) &= 0 \\
\sum_{i=1}^{n} x_{i2}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \ldots - \hat{\beta}_k x_{ik}) &= 0 \\
&\vdots \\
\sum_{i=1}^{n} x_{ik}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \ldots - \hat{\beta}_k x_{ik}) &= 0
\end{aligned}
$$

2. **How to do interpretation:** “Partialling out” interpretation of multiple regression

The Expected Value of the OLS Estimators

- **Expected Value:**

Unbiasedness of OLS (Chapter 3 Appendix)

In the model with *k* independent variables, we can write $\hat{\beta}_j$ as:

$$\hat{\beta}_j = \frac{\sum_{i=1}^{n}(\hat{r}_{ij}y_i)}{\sum_{i=1}^{n}(\hat{r}_{ij})^2} = \beta_j + \frac{\sum_{i=1}^{n}(\hat{r}_{ij}u_i)}{\sum_{i=1}^{n}(\hat{r}_{ij})^2}$$

where $\hat{r}_{ij}$ is the residual from a regression of x_j on $x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_k$

$$E(\hat{\beta}_j) = \beta_j + E[\frac{\sum_{i=1}^{n}(\hat{r}_{ij}u_i)}{\sum_{i=1}^{n}(\hat{r}_{ij})^2}]$$

$$= \beta_j + \frac{\sum_{i=1}^{n}(\hat{r}_{ij}E(u_i))}{\sum_{i=1}^{n}(\hat{r}_{ij})^2} = \beta_j$$

Under assumptions MLR.1 – MLR.5:

- **Variance:**

Variance of the error term

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{SST_j(1-R_j^2)}, \quad j = 1, \ldots, k$$

Total sample variation in explanatory variable x_j:

$$\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2$$

R-squared from a regression of explanatory variable x_j on all other independent variables (including a constant)